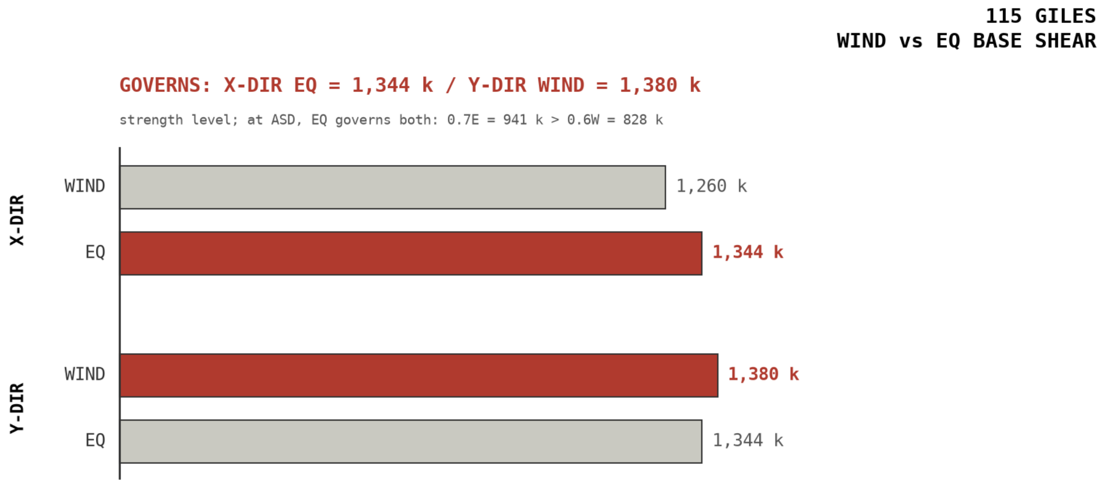
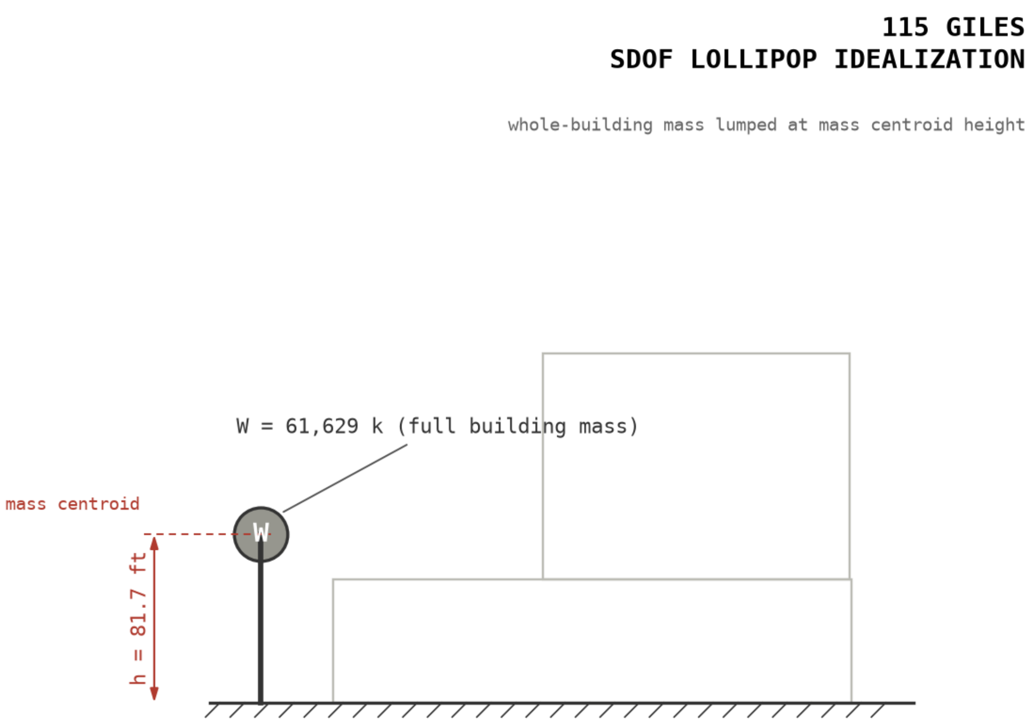
1 GOVERNING LATERAL LOAD



source: 115-GILES-PRELIM-LATERAL-LOADING (ASCE 7-16 case)

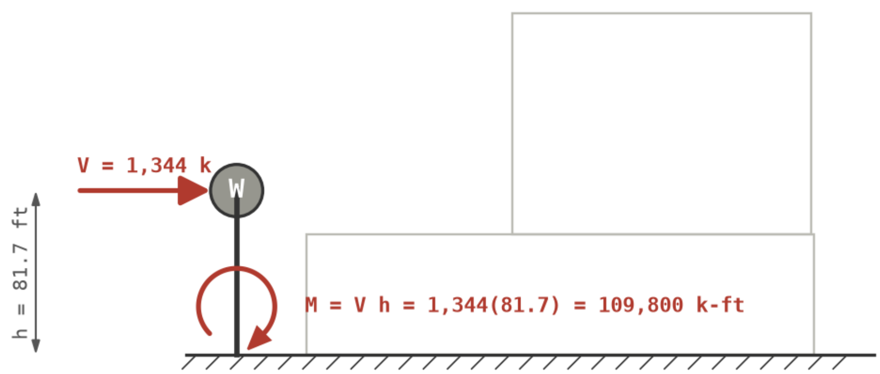
2 SDOF IDEALIZATION



3 EQ OVERTURNING MOMENT

115 GILES
EQ OVERTURNING MOMENT

EQ base shear applied at mass centroid of SDOF lollipop



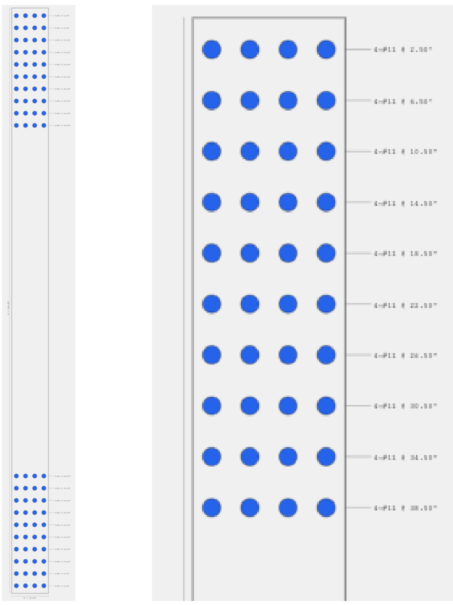
4 WALL QUANTITY

5 walls at 16 ft in X, 3 walls at 16 ft in Y. Each straight wall taken separately; torsion neglected. Demand split evenly; governing case per direction (X: EQ, Y: wind, resultant at $h/2 = 85$ ft).

$$\text{X:} \quad V_u = \tfrac{1344}{5} = 269 \text{ k} \qquad M_u = \tfrac{109{,}800}{5} = 21{,}960 \text{ k-ft} \tag{1}$$

$$\text{Y:} \quad V_u = \tfrac{1380}{3} = 460 \text{ k} \qquad M_u = \tfrac{1380(85)}{3} = 39{,}100 \text{ k-ft} \tag{2}$$

Section: 12 in $\times$ 16 ft, $f'_c = 5$ ksi, $f_y = 60$ ksi. Each end: 10 rows of 4#11 at 4 in. Section and flexural capacity from concrete-beam.vercel.app:



full wall / end zone

A_s (each end)	62.46 in^2
d / d'	171.5 / 20.5 in
a / c	28.4 / 35.5 in
ε_t	0.0115 (tension-controlled)
ϕM_n	43,124 k-ft

$$\phi V_n = \phi A_{cv}\,(2\lambda\sqrt{f'_c} + \rho_t f_y) = 0.75(2304)(141 + 150)/1000 = 504 \text{ k}$$

ε_t	net tensile strain, extreme tension steel
A_{cv}	wall shear area, $12 \times 192 = 2{,}304$ in^2
ρ_t	horizontal web reinf ratio, 0.0025 min assumed

$P = 0$ and equal end reinforcement; wall treated as vertical doubly-reinforced beam.

WALL	DCR V	DCR M
X (5 at 16 ft)	269/504 = 0.53	21,960/43,124 = 0.51
Y (3 at 16 ft)	460/504 = 0.91	39,100/43,124 = 0.91

Quantity works. Y walls at DCR ≈ 0.9 – tight; X has margin.